Binary · AS · Practice

U02a · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Unless a real-paper citation is shown, questions are ours. Values are unsigned. Where a question retains low bits, use exactly that stated model.

1. Ours: write the register

(a) Convert **91** to eight-bit binary. Show your working. **[2]**

(b) Convert **172** to eight-bit binary. Show your working. **[2]**

2. Ours: read the register

(a) Give the denary value of **10110011**. Show the place values you add. **[2]**

(b) Give the denary value of **01111110**. Show the place values you add. **[2]**

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3. Ours: capacity and the zero label

(a) For an eight-bit register, state the number of patterns and the largest unsigned denary value. **[2]**

(b) For a 12-bit register, state the number of patterns and the largest unsigned denary value. **[2]**

4. Ours: the carry that does not fit

(a) Add **10011010** and **01100110**. Show your carries and the full mathematical sum in binary. **[2]**

(b) This machine retains only the lowest eight bits of the sum. State the retained eight-bit pattern. **[1]**

(c) Explain why unsigned overflow occurs in this calculation. **[1]**

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5. Ours: exam-shape rehearsal · controller states

A controller uses a 10-bit unsigned state register. This is a fresh authored instance of the bit-pattern counting question shape.

(a) State the number of distinct binary patterns the 10-bit register can represent. **[1]**

6. Ours: a game score with an explicit rule

A score is 250 and the player earns 10 points. The storage model keeps only the lowest eight bits of the mathematical result.

(a) Show the mathematical total in denary and binary, with positional or addition working. **[2]**

(b) State the retained eight-bit pattern and its denary value. **[1]**

(c) Explain why the retained value differs from the mathematical total. **[1]**

7. Ours: grouping and representation

(a) State why binary can represent a switch with two states. **[1]**

(b) State the number of bits in a nibble. **[1]**

(c) State the number of bits in a byte. **[1]**

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8. Ours: six answers make a number · pen first [6]

💻 **Paper first, then the classroom cards.** No Python file is needed. Six cards are labelled by weights 1, 2, 4, 8, 16, 32. A number appears on a card exactly when that weight is used in its binary representation. Answer before trying the cards. These six marks are human marked.

(a) For the number **27**, write YES or NO in card order **1, 2, 4, 8, 16, 32**. Show the weights you used. **[2]**

(b) A player answers **NO, YES, NO, YES, NO, YES** in that same order. Write the six-bit pattern (largest weight first) and its denary value. **[2]**

(c) Add a seventh card. State its weight and the largest number the seven cards can identify when at least one answer is YES. **[2]**

Keyboard companion: continue Karel 05 LightTrail and 06 TheLongHall, your first loops. This companion is outside the 30 marks.

Above and beyond · outside the total. A counter has three faces per wheel: 0, 1, 2. Predict what follows 022. Explain the denary values of 022 and 100 using place values. Why does a carry happen at a different moment?

Additional space - label the question number.